

Boyang (Boe) Zhou

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Education

University of Washington, Seattle
M.S. in **Computer Science**, GPA 3.9/4.0

Anticipated 6/2026

University of Washington, Seattle
B.S. in **Computer Science**, Minor in **Environmental Studies**, Mary Gates Scholar, GPA 3.96/4.0

9/2021 – 3/2025

Research experience

Student Researcher - HCI, Accessibility, AR
Makeability Lab, University of Washington, Seattle, WA
Advised by **Prof. Jon E. Froehlich**

11/2024 – Present

- Leading a 3-person research team in exploring the design space of AI-powered always-on sensing for urban accessibility. Developed a multi-model phone-based sensing pipeline with a context-aware design, enabling always-on obstacle sensing to assist users in complex urban environments through adaptive visual, tactile, and audio feedback. The work is on track to be submitted to **UIST 2026**, where I am the first author.
- Computationally designed and fabricated a foldable iPhone attachment that redirects LiDAR sensing from downward-facing to forward-facing in a natural phone-holding position, significantly reducing user adoption barriers and enabling natural phone use while simultaneously detecting obstacles in the path.
- Conducting a user study integrating qualitative interviews with field metrics to assess deployability. Utilized real-time gaze data as a proxy for analyzing behavioral shifts and safety during urban navigation.
- Fine-tuned open-vocabulary visual segmentation models on a SoccerNet dataset and built a serialized inference pipeline to run locally on the Quest VR headset. Achieved on-device inference to power a visual augmentation system for low-vision individuals to experience live soccer events.

Student Researcher - Graphics, Diffusion, 3D/4D Reconstruction
Reality Lab, University of Washington, Seattle, WA
Advised by **Prof. Steven Seitz** and **Prof. Ira Kemelmacher-Shlizerman**

6/2022 – Present

- Co-leading the development of a 4D Gaussian Splatting system that stabilizes monocular reconstructions using explicit deformations and curriculum flow-matching, producing artifact-free 6DoF VR experiences from casual videos; on track for **SIGGRAPH 2026** submission.
- Developed a novel joint fine-tuning pipeline for the attention layers of video diffusion models to enable temporally consistent generative frame interpolation in animation workflows, reducing training cost by fine-tuning only 2% of model parameters while preserving motion expressiveness; work published at **ICLR 2025**.
- Curated a motion-rich video dataset and orchestrated large-scale training experiments across HPC GPU clusters to evaluate model performance under varying dataset and architectural configurations.
- Developed a pipeline for Finite Element Analysis simulation based on lumbar models reconstructed from MRI scans. The model was used in an AR-based lumbar injection training simulator on Microsoft HoloLens, enabling a prototype of radiation-free Epidural Steroid Injection procedure. The work is presented at the Annual Meeting of the American Society of Pain and Neuroscience.

Student Researcher - NLP
Climate Risk Lab, University of Washington, Seattle, WA

11/2022 – 6/2023

- Developed an NLP-driven pipeline leveraging sentiment analysis and keyword extraction to evaluate TCFD report quality and climate risk exposure, increasing processing speed by 30× over human benchmarks.

Working Papers

Lucas Wang, Marianny De León, **Boyang Zhou**, Sai Jayanth Kalisi, Keith Meng Phou, Jennifer Han, Yijun Chen, Alicia Betty, Oliver Aalami, Alberto Paderno, Rania Hussein. 2025 (in press). CARE - Clinician Augmented Reality Environment: Design and Technical Feasibility of Apple Vision Pro for Image-guided Surgeries. *Journal of the American Medical Informatics Association (JAMIA)*. [Link]

Baback Elmieh, **Boyang Zhou**, Steve Seitz, Brian Curless, Ira Kemelmacher. 2026. Dynamic 6DoF VR reconstruction from monocular videos. In preparation for submission.

Boyang Zhou*, Oleg Ianchenko*.(^{*}equal contribution) 2026. Addressing Negative Commons Governance with Positive Commons Principles. In preparation for submission.

Haijing Zong*, **Boyang Zhou***, Yancheng Liang, Rachel Heath, Natasha Jaques.(^{*}equal contribution) 2026. Multi-agent Reinforcement Learning benchmark for job market search and matching. In preparation for submission.

Boyang Zhou, Biira B, Kurtis Heimerl. 2026. How Does Machine Learning Community Discuss Ethics? An Analysis of NeurIPS Papers' Ethics Statements.

Publications

Xiaojuan Wang, **Boyang Zhou**, Brian Curless, Ira Kemelmacher, Aleksander Holynski, Steve Seitz. 2025. Generative Inbetweening: Adapting Image-to-Video Models for Keyframe Interpolation. In *Proceedings of the International Conference on Learning Representations (ICLR 2025)*. [Link]

Boyang Zhou, Johan Lindqvist, Lindsey Li. 2025. Reproducibility Report: Test-Time Training on Nearest Neighbors for Large Language Models. In *arXiv 2511.16691*. [Link]

Work experience

Grant and Project Coordinator (20hr/wk) 6/2023 – 6/2025

University of Washington Campus Sustainability Fund, Seattle, WA

- Directed a portfolio of over 50 campus sustainability projects (solar, biodiversity, environmental justice, etc.) funded by a \$500,000/year grant. Practiced justice-centered grant making and shared work at multiple regional/national conferences. [Link to project portfolio]
- Contributed to several campus-as-a-living-lab research projects, focusing on developing computing tools to solve real-life ecological issues such as a mobile app for crowdsourcing bird-building collision data, and quantifying the amount of tire-derived toxic chemicals in rainwater runoff.
- Served as the primary liaison between project teams and diverse stakeholders, including tribal representatives, local nonprofits, infrastructure engineers, and research groups. Conducted over 300 meetings.
- Helped implement participatory budgeting and decision-making practices with an eight-member student committee at Campus Sustainability Fund, becoming the first organization at the university to establish such a governance structure.

Software Engineer Intern - GIS (40hr/wk) 6/2024 – 12/2024

King County Department of Natural Resources & Parks, Seattle, WA

- Engineered and deployed new features for Lucity GIS, integrating disparate geospatial (ArcGIS) and asset PostgreSQL databases through gRPC. Streamlined infrastructure maintenance workflows, resulting in a significant reduction in manual data entry for 350+ park maintenance crews, rangers, and environmental scientists.
- Redesigned mobile data logging interfaces to support geo-tagged visual documentation (images/video), enabling rangers to replace text-heavy logs with rich visual evidence.
- Developed Python and ArcGIS-based data analytics pipelines to analyze county demographic and environmental data, identifying five underserved communities and informing updates to the trail master plan for equitable transportation infrastructure development. Presented the trail master plan proposal to executives, which was subsequently adopted into the 2025 King County Park Levy.

AI Software Engineer Intern (20hr/wk) 6/2023 – 9/2023

Hugsi, Bellevue, WA

- Architected an error-aware RAG pipeline using GPT-3.5 for a Flutter-based iOS app, improving the relevance of personalized gift suggestions on the homepage and reducing hallucinations by 65% based on internal evaluations.

Leadership & Teaching experience

Teaching Assistant (CSE557: Computer Graphics & CSE447: NLP) 9/2025 – Present
University of Washington, Seattle, WA

- Developed and extended a C++ codebase for CMU's Scotty3D graphic engine used by 20 master's and Ph.D. students, adding instructional features such as improved test cases, segfault debugging tools, auto-save functionality, and enhanced accessibility of course materials.

XR Club President 11/2022 – 9/2025
Extended Reality Association at the University of Washington, Seattle, WA

- Led a team of seven officers and over 80 students to build an interdisciplinary Extended Reality (AR/VR) community in the Seattle area through hosting technical workshops, speaker series, and hackathons.
- Secured a \$60,000 grant to purchase Vision Pros and Quest headsets accessible to all students.
- Utilized LiDAR scanning and point cloud processing to digitally reconstruct high-accuracy accessibility maps for two major campus buildings, delivering actionable data to the building management and visitor center.

Summer Camp Counselor 6/2022 – 9/2022
iD Tech Camp, Seattle, WA

- Taught over 100 students (ages 7 to 17) about JavaScript programming, game development in Unreal Engine, and other technical lessons, with a 92% student satisfaction rate

Talks

"Measuring the impact of student-led sustainability" (with Campus Sustainability Fund) 2024
Association for the Advancement of Sustainability in Higher Education Conference

"Leading justice-centered sustainability" (with Campus Sustainability Fund) 2023, 2024
Washington Oregon Higher Education Sustainability Conference

Volunteering

TorchGeo open-source contributor 2025

UW Farm volunteer 2023-2025

Washington State Congressional App Challenge mentor 2024, 2025

SIGGRAPH student volunteer 2024

AWE student volunteer 2024

Sierra Club Northwest Forest Plan coordinator 2024

Fellowships & Awards

Mary Gates Scholarship 2024
\$5,000

Stanford XR Hackathon "Best Use of AI" 2023
\$500

Skills

Programming Languages: Java, Python, Rust, C++, TypeScript, Swift, R, C#, GLSL, SQL

Methods: Natural Language Processing, Reinforcement Learning, Human-Computer Interaction, Fabrication, Neural Rendering, Gaussian Splatting, Diffusion, Flow Matching, 3D Printing, Extended Reality, Geospatial Machine Learning

Tools: Slurm, PyTorch, JAX, Google Cloud, AWS, Docker, Apptainer, React, Apache Spark, PostgreSQL, OpenXR, RealityKit, Unity, Unreal, Three.js, ArcGIS, MCP, LlamaIndex, Blender, SolidWorks

Other: Grant writing, Literature reviews, Structured & Semi-structured interviews, Survey design, Project management